My health @ EU
eHealth Digital Service Infrastructure
A service provided by the European Union

**My Health @ EU**
**eHealth Digital Service Infrastructure**
**Electronic cross-border health services in the EU**

# XCA Profile

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| ABSTRACT |
|---|
| This document specifies the binding of the eHealth DSI "Fetch Document" service to the IHE XCA profile. |


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| | | | | |
|---|---|---|---|---|
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| V2.0.1 | 19/05/2017 | Update error message codes (1101 or 1102) when no data | eHealth DSI provider | |
| V2.0.0 | 28/03/2017 | Remove all references to epSOS and requirements | eHealth DSI provider | |
| V1.0 | 17/12/2013 | Integration of comments | Fraunhofer FOKUS | |

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# 1 Introduction

This document specifies the binding of the eHealth DSI "Fetch Document" service to the IHE XCA profile.

## 1.1 eHealth DSI Country-B Document Request

Through the "Fetch Document" interaction pattern eHealth DSI provides functionalities to a country-B-data-consumer that allow him to retrieve medical data about patient-A from country A. The eHealth DSI "Fetch Document" Service maps this interaction pattern onto message exchanges within the eHealth DSI domain (see figure 1).

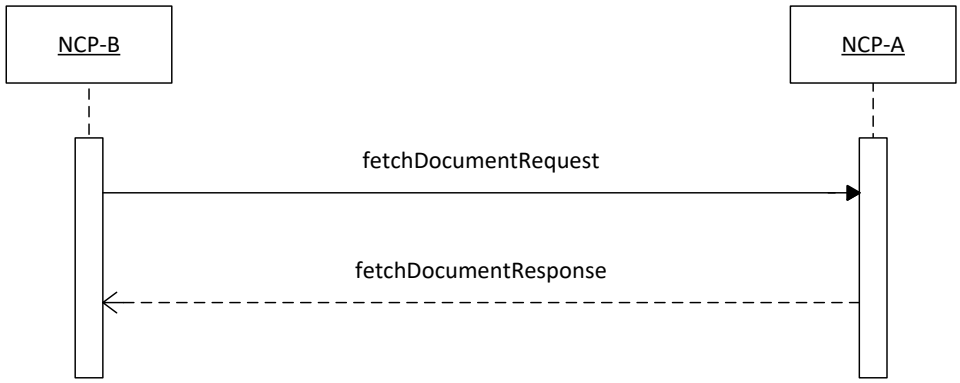



**Figure 1: eHealth DSI Fetch Document Transactions**

The service is always implemented at NCP-A and requested by NCP-B. The service contract consists of two messages: *fetchDocumentRequest* and *fetchDocumentResponse*.

In order to query and retrieve certain data of an identified patient NCP-B MUST first send a *fetchDocumentRequest* message to NCP-A. This message carries information to discover the requested data and an optional indicator on the requested data format(s). NCP-A MUST respond to a f*etchDocumentRequest* message by sending back a *fetchDocumentResponse* message to the requestor. This message carries the fetched documents and (in case documents are provided in multiple formats) information about the relationship among these documents.

## 1.2 eHealth DSI Fetch Document Service XCA Binding

The eHealth DSI specification [Interoperability Specification] defines the logical interfaces, the behaviour, and the overall semantics of the Fetch Document Service but does not mandate for a specific technical standard for implementing the fetchDocument request and response messages. This document specifies a respective binding onto the IHE Cross-Community Access (XCA) integration profile.

### 1.2.1 Profile Overview

In [IHE ITI TF-1] the functionality and semantics of the XCA integration profile is defined as:

*The Cross-Community Access profile supports the means to query and retrieve patient relevant medical data held by other communities. A community is defined as a coupling of facilities/enterprises that have agreed to work together using a common set of policies for the purpose of sharing clinical information via an established mechanism. Facilities/enterprises may host any type of healthcare application such as EHR, PHR, etc.*

In the eHealth context the messaging infrastructure and security framework of the NCP-to-NCP eHealth DSI domain provide the reliable messaging platform to XCA. Within the eHealth DSI domain NCP-B takes the role of the document consumer while

NCP-A acts as a proxy for the document source within country-A's national infrastructure.

### 1.2.2 eHealth DSI Profiling of the XCA Integration Profile

Within eHealth DSI the actors and transactions of the IHE XCA integration profile are mapped onto eHealth DSI services and messages as follows:

| **IHE XCA** | **eHealth DSI Fetch Document Service** |
|---|---|
| Actor: Initiating Gateway | NCP-B Fetch Document Service consumer |
| Actor: Responding Gateway | NCP-A Fetch Document Service provider |
| Transaction: Cross-Gateway Query | fetchData request and response messages implemented as a sequence of the two IHE XCA transactions (see section 1.2.3) |
| Transaction: Cross-Gateway Retrieve | |

This eHealth DSI binding introduces minor extensions and restrictions on the IHE XCA actor and transaction definitions in order to properly cover the eHealth DSI use cases and to align with the eHealth DSI security framework:

- Additional error messages are defined that cover specific failure conditions of the eHealth DSI use cases
- The optionality of data fields is aligned to the specific eHealth DSI privacy requirements that may further constrain the availability and/or applicability of (meta) data

The application of security measures and the contents of the SOAP security header are specified normatively (see [Messaging Profile]).

### 1.2.3 eHealth DSI Sequencing of XCA Transactions (epSOS-2)

As defined in the functional specification, the Fetch Document Services allows for retrieving patient data by their semantic signifier. The following figure sketches how XCA with *FindDocuments* query semantics is used for fetching eHealth DSI data for a given semantic signifier. The respective binding is specified in chapter 2 of this specification document.

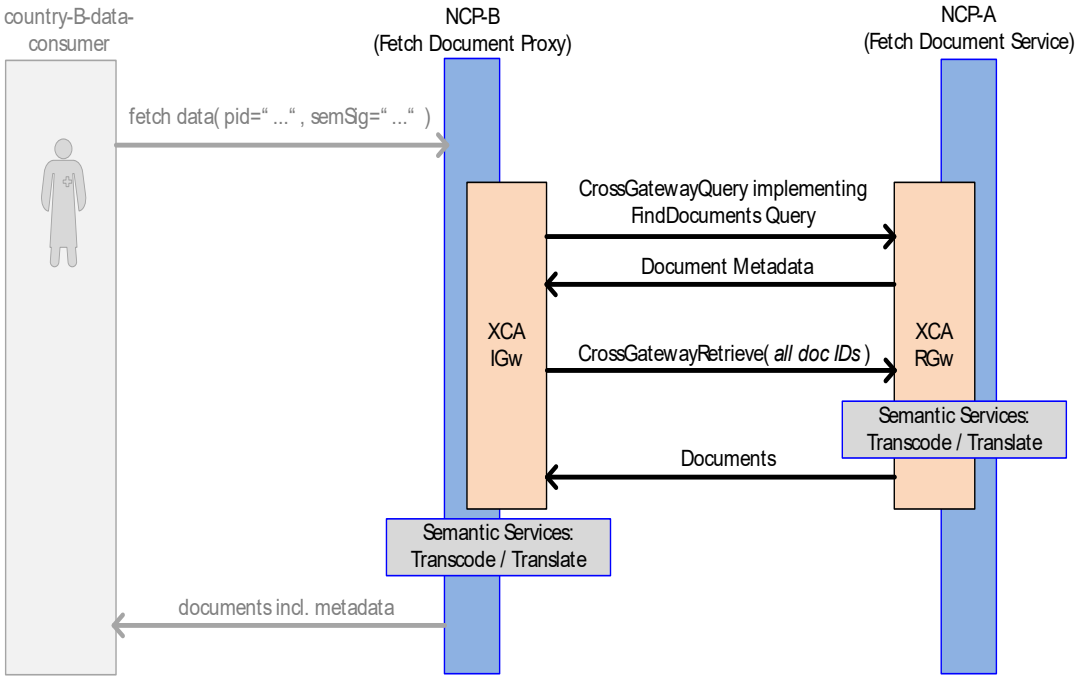



**Figure 2: Sequencing of IHE XCA Transactions**

### 1.2.4 epSOS-1 Compatibility Profile for Overview Documents

While the split query-retrieve transaction of XCA works rather efficient for the retrieval of single documents (e.g. a patient summary) there were performance issues reported from epSOS-1 pilots when it comes to the sharing of larger document sets (e.g. ePrescriptions). For this reason epSOS-2 introduced the concept of "overview documents" which encode a table of content over such document sets and allow for a healthcare professional at country-B to select among this table the documents to be retrieved.

It must be assumed that not all recently running epSOS-1 compliant NCPs will provide support for such overview documents. This is especially an issue in cases where an epSOS-2 enabled NCP-B wants to interact with an epSOS-1 NCP-A, because in this case NCP-B cannot provide the full epSOS-2 functionality to HPs in country B. To solve this issue, this binding specifies a "epSOS-1 Compatibility Profile for Overview Documents" that allows for NCP-B to implement full epSOS-2 functionality to its clients even in cases where the cooperating gateway (NCP-A) does not support epSOS-2 overview documents.

The compatibility profile enables an NCP-B to assemble an overview document solely from the document set metadata provided from an epSOS-1 NCP-A. It then provides this document to HP-B who selects among the set the documents to be retrieved. For all selected documents a document retrieval transaction is send to NCP-A which then provides the requested documents (see XCA Binding for the epSOS-2 "Retrieve Document Service"). Figure 3 sketches the flow of control for this binding

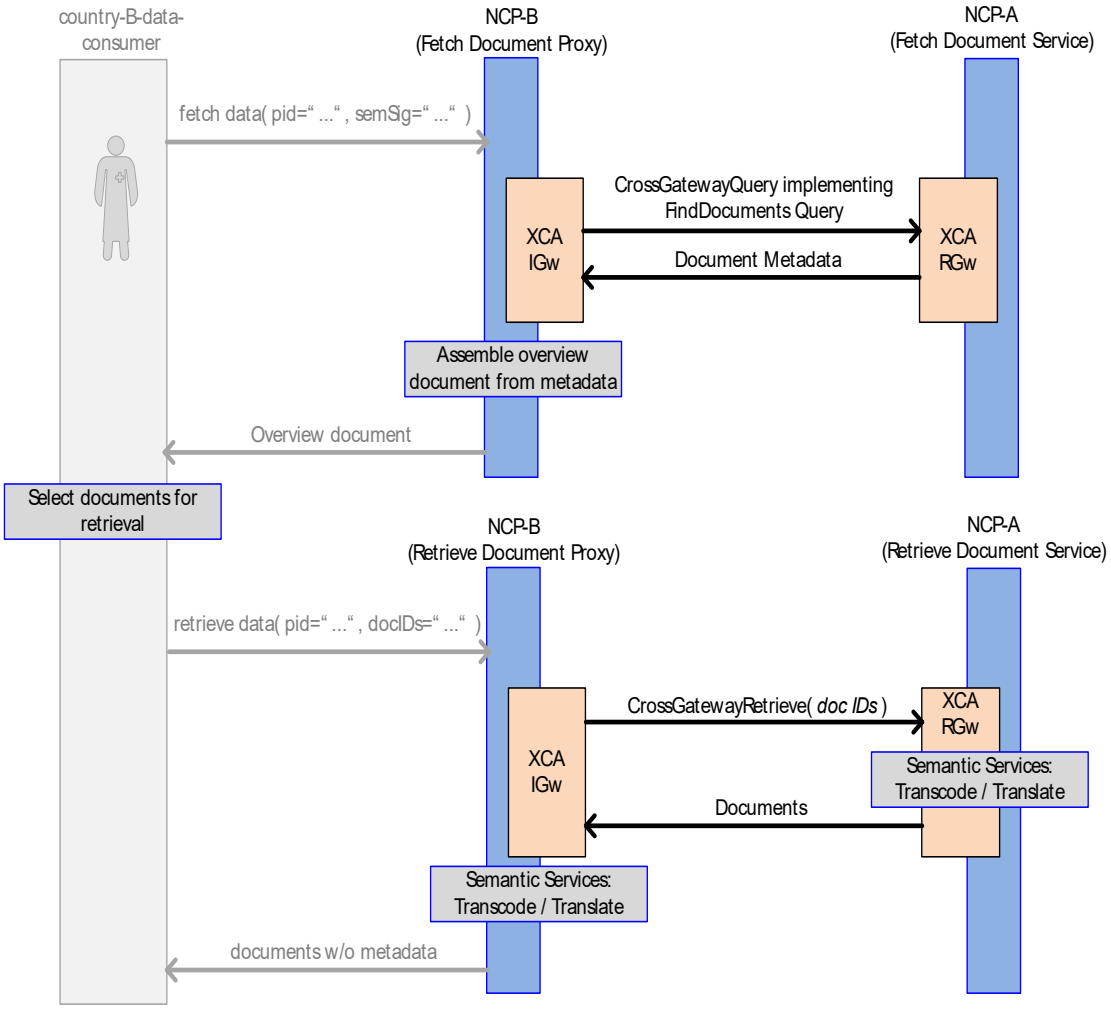



**Figure 1: Implementing Overview Documents Using the eHealth DSI XCA Profile**

For this interoperability profile to work, NCP-A

- MUST have implemented XCA Cross-Gateway Query and XCA Cross-Gateway Retrieve as independent transactions

- SHOULD provide additional information to be displayed to HP-B within the document metadata provided by a Cross-Gateway Query for a document set (this is already supported by some of the NCPs which are based on the OSS NCP)

- MAY implement the XCA On Demand Document option (which means that the compatibility profile is robust for static as well as for dynamic documents).

The respective binding is specified in chapter 3 of this specification document.

## 1.3 Related Documents

The functional specification of the Fetch Data Service is defined in [Interoperability Specification] which is the normative foundation for all binding specifications. The technical binding defined in this document makes direct use of other bindings:

- [Audit Trail Profiles] for a specification of the eHealth DSI common audit trail contents and structure
- [Messaging Profile] for a specification of how SOAP and Web Service Security are to be used for implementing eHealth DSI services
- [SAML Profile] for a specification of the HP Identity Assertion and the Treatment Relationship Conformation Assertion which are used for providing authentic information about the consumer of an eHealth DSI service and the recent care context

The medical documents to be transmitted by the Fetch Data service are specified through Semantic Signifiers (see [Interoperability Specification]). Examples of such signifiers are:

- ePrescription and ePrescription Overview for accessing dispensable medications
- Patient Summary for reporting on a patient's health status

## 1.4 Conventions

The keywords MUST, SHOULD, MAY, SHOULD NOT and MUST NOT are used as defined in [RFC 2119].
eHealth DSI requirements are managed within a central requirements management tool and serialized into [Requirements and Recommendations]. References to requirements always use the identifiers as used with the central requirements. Newly defined requirements are handed over to the central requirements management. In case of doubt or conflict, [Requirements and Recommendations] MUST be considered to be the normative phrasing of a requirement.

## 1.5 Terms and Definitions

The country which holds information about a patient, where the patient can be univocally identified and where his data may be accessed is called "country-A" (country of affiliation) [eHealth DSI Glossary]. The country of treatment where cross-border health care is provided when the patient is seeking care abroad is called "country-B" (country of care) [eHealth DSI Glossary].
The term "health professional (HP)" denotes a doctor of medicine, a nurse responsible for general care, a dental practitioner, a midwife or a pharmacist within the meaning of Directive 2005/36/EC, or another professional exercising activities in the healthcare sector which are restricted to a regulated profession as defined in Article 3(1)(a) of Directive 2005/36/EC, or a person considered to be a health professional according to the legislation of the country of treatment [eHealth DSI Glossary]. Health professionals are allowed to process medical patient data according to the legislation of the country

of the health professional's residence. An "eHealth DSI Point of Care (PoC)" is a location where an eHealth DSI patient may seek healthcare services [eHealth DSI Glossary].
A "National Contact Point (NCP)" is an entity in each particular country to act as a bidirectional technical, organizational and legal interface between the existing national functions and infrastructures [eHealth DSI Glossary]. In this document the term "NCP" emphasizes the technical aspects of this interface and as such refers to a gateway which facilitates various aspects of cross-border data sharing (e.g. message forwarding, signature verification). From an architectural perspective NCPs denote the boundary between the eHealth DSI infrastructure and a country's existing national eHealth infrastructure (see [Specifications] for details).

## 1.6 Namespaces

XML namespace prefixes are used in this document to stand for their respective namespaces as follows.

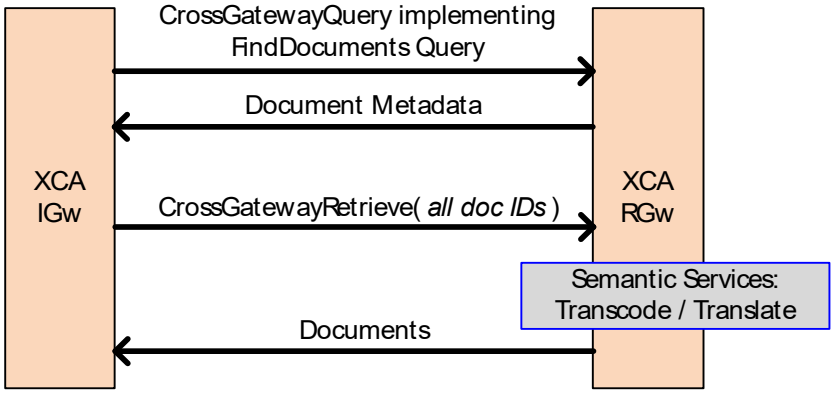
| Prefix | Namespace |
|---|---|
| ehdsi | urn:ehdsi |
| soapenv | http://www.w3.org/2003/05/soap-envelope |
| wsse | http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1.0.xsd |
| saml | urn:oasis:names:tc:SAML:1.0:assertion |
| xacml | urn:oasis:names:tc:xacml:2.0:policy:schema:os |
| hl7v2 | urn:hl7-org:v2 |
| hl7v3 | urn:hl7-org:v3 |
| xs | http://www.w3.org/2001/XMLSchema |
| xsi | http://www.w3.org/2001/XMLSchema-instance |
| query | urn:oasis:names:tc:ebxml-regrep:xsd:query:3.0 |
| rim | urn:oasis:names:tc:ebxml-regrep:xsd:rim:3.0 |
| rs | urn:oasis:names:tc:ebxml-regrep:xsd:rs:3.0 |
| lcm | urn:oasis:names:tc:ebxml-regrep:xsd:lcm:3.0 |

## 1.7 Status of this Binding

The binding as defined in this document is a normative binding. All eHealth DSI countries MUST implement this binding within their NCP.
Other eHealth DSI documents must refer to this binding as [XCA Profile].
Additional or alternative bindings for the *eHealth DSI Document Fetch Service* MUST NOT be defined and implemented for the eHealth DSI.

# 2 eHealth DSI Fetch Document Service

The *eHealth DSI Fetch Document Service* request and response messages are implemented by an XCA CrossGatewayQuery transaction (ITI-38) followed by an XCA CrossGatewayRetrieve transaction (ITI-39). The CrossGatewayQuery follows the query syntax and semantics of the ITI-18 FindDocuments query.




## 2.1 FindDocuments Request Message

The FindDocuments message is a specialisation of the *IHE CrossGatewayQuery* transaction (ITI-38) request message as profiled in [IHE ITI TF-ITI-38]. The query arguments defined for this message MUST follow the syntax and semantics of an ITI-18 FindDocuments query as profiled in [IHE ITI TF-2]. As a restriction only the query parameters listed in the following table SHOULD be used. Other parameters MAY be used but MAY be ignored by NCP-A.

| Parameter | Opt. | Usage Convention |
|---|---|---|
| $XDSDocumentEntryPatientId | R | MUST provide the HL7v3 II coded patient identifier previously obtained by the eHealth DSI Patient Identification Service. |
| $XDSDocumentEntryClassCode | R | See section 2.1.1 |
| $XDSDocumentEntryFormatCode | O | The default document format is defined by the Semantic Signifier. If the Semantic Signifier allows to overwrite this default or to explicitly determine a requested format, this argument MAY provide an URN that denotes whether eHealth DSI pivot coded documents or PDF/A source coded documents are requested. |
| $XDSDocumentEntryStatus | R | MUST be 'urn:oasis:names:tc:ebxml-regrep: StatusType:Approved' |

### 2.1.1 Semantic Signifiers

epSOS-2 introduces the concept of a "Semantic Signifier" as a holistic specification and usage guidance for document classes. A semantic signifier is an XML structure that encapsulates conceptual models for defining the semantics of a document class, formal specifications (e.g. as schemas and Schematrons) for implementing and verifying respective data objects, and approved transformation scripts (e.g. for visualization).

In order to preserve compatibility with [IHE ITI TF-3] XCA and eHDSI decouples the semantic signifier structure from the IHE X* metadata by using the LOINC-coded document class code as an indirect reference to the respective semantic signifier:

- classCodes MUST be coded according to specification in [ITI TF-2a] section 3.18.4.1.2.3.4 Coding of Code/Code-Scheme
- For classCodes the classification scheme 2.16.840.1.113883.6.1 with codeSystemName="LOINC" MUST be used
- For document classes (ePrescription, eDispensation, Patient Summary, OrCD Laboratory Result, OrCD Hospital Discharge Report, OrCD Medical Imaging Report, OrCD Medical Images, Patient Consent) the LOINC codes as defined in [CDA templates] MUST be used for the respective document classes

eHealth DSI Semantic Signifiers MUST be univocally mapped on LOINC Codes in order to allow an NCP to obtain a document class Semantic Signifier from the document class coded that is coded within an IHE X* message.

### 2.1.2 Use of Pseudonyms and Temporal Identifiers

eHealth DSI MUST NOT make any assumptions on how a requested document is assembled within country-A. This allows as well for pre-rendered documents as for the on-demand rendering upon request. While such considerations are almost irrelevant on the functional level, the IHE XCA profile provides a dedicated "On-Demand Documents" option for the second scenario (which should be assumed as the default for most of the eHealth DSI countries). Therefore the IHE XCA binding to the "Fetch Documents" service MUST be aware of such country-A implementation details.
In order to be compliant with the XCA profile specification and to be downward compatible with existing epSOS-1 implementations, the following behaviour of the service consumer and provider is assumed as the default which MAY be overwritten by a semantic signifier for a specific class of documents:

| **Initiating Gateway** | **Responding Gateway** |
|---|---|
| The gateway does not implement the On Demand Documents option. This is the epSOS-1 assumption.<br>The query is only on static documents. | The gateway does not implement the On Demand Documents option. This is the epSOS-1 assumption.<br>The gateway MUST respond with registry metadata for all documents that provide the minimum set of documents that fulfil the semantics of the query as defined by the respective semantic signifier (e.g. most current patient summary document or most current set of active, dispensable ePrescriptions). |
| | The gateway implements the On Demand Documents option.<br>For epSOS-1 compatibility the gateway MUST act as described for the case where the On Demand Documents option is not implemented. |
| The gateway implements the On Demand Documents option.<br>The query is only on dynamic documents.<br>If the responding gateway responds with an error signalling that the requested option is not supported, the initiating gateway SHOULD re-send the query by requesting only static documents (see above). | The gateway does not implement the On Demand Documents option. This is the epSOS-1 assumption.<br>The gateway MUST either<br>• Ignore the query for dynamic documents and act as if the query was for static documents (see above).<br>• Respond with an error signalling that the requested option is not supported. |
| | The gateway implements the On Demand Documents option.<br>The gateway MUST respond with registry metadata for all documents that provide the minimum set of documents that fulfil the semantics of the query as defined by the respective semantic signifier (e.g. most current patient summary document or most current set of active, dispensable ePrescriptions). |
| The gateway implements the On Demand Documents option.<br>The query is on both static and dynamic documents. | The gateway does not implement the On Demand Documents option. This is the epSOS-1 assumption.<br>The gateway MUST ignore the On Demand Documents option and act as if the query was for static documents only (see above). |
| | The gateway implements the On Demand Documents option.<br>The gateway MUST at least respond with registry metadata for all documents that provide the minimum set of |

| | |
|---|---|
| | documents that fulfil the semantics of the query as defined by the respective semantic signifier (e.g. most current patient summary document or most current set of active, dispensable ePrescriptions).<br><br>Additionally it MAY include registry metadata for previously rendered instances of the requested documents with the response. The relationships among all instances/versions of a document MUST be expressed by including respective association entries within the response message. |

### 2.1.3 Example FindDocuments Request Message

See [Technical Annexes]

## 2.2 FindDocuments Response Message

The eHealth DSI response message to a *FindDocuments* query MUST implement the respective [IHE ITI TF-ITI-38] XCA *CrossGatewayQuery* response message. The response message MUST contain registry metadata for each document that matches the given semantic signifier and the given format. The following constraints to the metadata definitions apply:

| Slot Name | Binding | Slot Value |
|---|---|---|
| mimeType | Attribute | MUST be "text/xml" |
| sourcePatientId | rim:Slot | MUST be of the same value as $XDSDocumentEntry.PatientId |
| healthcareFacilityTypeCode | classification | MUST be provided for XCA compatibility and correct addressing. Value MUST be set to ISO 3166-1 alpha-2 country code of the addressed PN. |
| classCode | classification | See section 2.1.1 |
| formatCode | classification | Format qualifier as defined in table 1C of [CDA templates]; |
| patientId | External identifier | Equals to the patient identifier that was provided by the *eHealth DSI Identification Service* (encoded as HL7 v3 II data type) |

### 2.2.1 Associations

An ebRIM association MUST be used for declaring the eHealth DSI pivot coded document as a transformation of the source coded document. As classification scheme urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3 MUST be used per [IHE ITI TF-2] XCA. Currently "eHealth DSI pivot" is defined as the only valid transformation:

```
<rim:Association id="id of the association" associationType="urn:ihe:iti:2007:AssociationType:XFRM"
        sourceObject="UUID of the source coded document" targetObject="UUID of the eHealth DSI
        pivot document" objectType="urn:oasis:names:tc:ebxml-
        regrep:ObjectType:RegistryObject:Classification">
    <rim:Classification id="id of the classification" classificationScheme="urn:uuid:abd807a3-4432-4053-
87b4-fd82c643d1f3" classifiedObject="id of the association" objectType="urn:oasis:names:tc:ebxml-
regrep:ObjectType:RegistryObject:Classification" nodeRepresentation="eHealth DSI pivot">
        <rim:Slot name="codingScheme">
            <rim:ValueList>
                <rim:Value>eHealth DSI translation types</rim:Value>
            </rim:ValueList>
        </rim:Slot>
        <rim:Name>
            <rim:LocalizedString value="Translation into eHealth DSI pivot format"/>
        </rim:Name>
    </rim:Classification>
</rim:Association>
```

| Metadata (ebRIM names) | Binding | Health DSI Opt. | eHealth DSI usage convention |
|---|---|---|---|
| associationType | Attribute | R | MUST be "urn:ihe:iti:2007:AssociationType:XFRM" |
| sourceObject | Attribute | R | MUST be UUID of the source coded document |
| targetObject | Attribute | R | MUST be UUID of the eHealth DSI pivot document |
| Association Documentation | Main | O | MAY be empty. MAY be ignored by the service consumer. MAY contain a description of the Association according to the Association Documentation object as used by IHE. |
| classificationSheme | classification | O | MUST be "urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3" |
| classifiedObject | Attribute | O | MUST be ID of the association |
| nodeRepresentation | Attribute | O | MUST be "eHealth DSI pivot" |
| codingScheme | rim:Slot | O | MUST be "eHealth DSI translation types" and "Translation into eHealth DSI pivot format" |

### 2.2.2 Metadata

The following metadata can be provided for every document retrieved using the FindDocuments service:

| Slot name | Health DSI Opt. | Description | Example |
|---|---|---|---|
| size | O | Size in bytes of the byte stream that comprises the document.<br>In the case of an On Demand document generation, no information on the size is available at the time of the XCA List. | <Slot name="size"><br><ValueList><br><Value>5240</Value><br></ValueList><br></Slot> |
| hash | O | Document hash calculated with SHA1 algorithm.<br>In the case of an On Demand document generation, no information on the hash is available at the time of the XCA List. | <Slot name="hash"><br><ValueList><br><Value>2264d7f11d4c21f3<br>fd4d8d093a842d765009ce72<br></Value><br></ValueList><br></Slot> |
| creationTime | O | Creation date of the document in the date format **yyyyMMddHHmmss** | <Slot name="creationTime"><br><ValueList><br><Value>20230321121310<br></Value><br></ValueList><br></Slot> |

### 2.2.2.1 ePList Metadata

For the ePrescription service, the FindDocuments response message must be provided with the following metadata elements for every returned document:

| Data element | Health DSI Opt. | Description |
|---|---|---|
| Description | R | Description of the document |
| Dispensable | R | A flag indicating whether the prescription is "available" or "current", i.e. whether it can be made available for dispensation in Country B. |
| ATC code | O | Anatomical Therapeutic Chemical (ATC) classification of the prescribed medicinal product or the ATC code on which the generic prescription is directly based. |
| Strength | O | The strength of the prescribed medicinal product. |
| Dose form | O | Dose form of the prescribed medicinal product. |
| Prescriber | O | Name of the prescriber. |
| Substitution | O | A code indicating whether generic or therapeutical substitution is allowed. |

#### 2.2.2.1.1 Examples

##### *2.2.2.1.1.1 Description*

```
<Description>
    <LocalizedString value="FORSTEO 20MCG/80MCL SOL INY 1PLUM PREC 3ML"/>
</Description>
```

##### *2.2.2.1.1.2 Dispensable*

| | | |
|---|---|---|
| **classificationCode** | urn:uuid:2c6b8cb7-8b2a-4051-b291-b1ae6a575ef4 | |
| **nodeRepresentation** | Yes | urn:ihe:iti:xdw:2011:eventCode:open |
| | No | urn:ihe:iti:xdw:2011:eventCode:closed |
| **Code System OID** | 1.3.6.1.4.1.19376.1.2.3 | |
| **Value** | Yes | Open |
| | No | Closed |

```
<Classification classificationScheme="urn:uuid:2c6b8cb7-8b2a-4051-b291-
b1ae6a575ef4"
classifiedObject="urn:uuid:5e4f66e0-9571-43be-a48a-9f786e4a0891"
id="urn:uuid:33b4750b-85e9-452d-bbee-080bc1173424"
nodeRepresentation="urn:ihe:iti:xdw:2011:eventCode:open">
   <Slot name="codingScheme">
        <ValueList>
             <Value>1.3.6.1.4.1.19376.1.2.3</Value>
        </ValueList>
   </Slot>
   <Name>
        <LocalizedString value="Open"/>
   </Name>
</Classification>
```

#### *2.2.2.1.1.3 ATC Code*

| | |
|---|---|
| **classificationCode** | urn:uuid:2c6b8cb7-8b2a-4051-b291-b1ae6a575ef4 |
| **nodeRepresentation** | ATC code |
| **Code System OID** | 2.16.840.1.113883.6.73 |
| **Value** | Display name for the ATC code |

```
<Classification classificationScheme="urn:uuid:2c6b8cb7-8b2a-4051-b291-
b1ae6a575ef4"
classifiedObject="urn:uuid:b527bd29-569d-4c61-83d1-ac23ccab9608"
id="urn:uuid:864f3c70-7c17-498f-b825-d51a0de6c0d8"
nodeRepresentation="C10BX06">
    <Slot name="codingScheme">
        <ValueList>
             <Value>2.16.840.1.113883.6.73</Value>
        </ValueList>
    </Slot>
    <Name>
        <LocalizedString value="C10BX06 - Atorvastatina, acido acetilsalicil
ico y ramipril"/>
    </Name>
 </Classification>
```

#### *2.2.2.1.1.4 Strength*

| | |
|---|---|
| **classificationCode** | urn:uuid:2c6b8cb7-8b2a-4051-b291-b1ae6a575ef4 |
| **nodeRepresentation** | Strength of the medicinal product |
| **Code System OID** | eHDSI_Strength_CodeSystem |
| **Value** | Strength of medication |

```
<Classification classificationScheme="urn:uuid:2c6b8cb7-8b2a-4051-b291-
b1ae6a575ef4"
classifiedObject="urn:uuid:343b5a5b-bd3a-49a3-989f-edc1970e059b"
id="urn:uuid:2df46b4c-77aa-4cb5-aba3-fce07d363cb5"
nodeRepresentation="20 mg/L">
   <Slot name="codingScheme">
        <ValueList>
             <Value>eHDSI_Strength_CodeSystem</Value>
        </ValueList>
   </Slot>
```

```
   <Name>
        <LocalizedString value="Strength of medication"/>
    </Name>
</Classification>
```

#### 2.2.2.1.1.5 Dose form

| | |
|---|---|
| **classificationCode** | urn:uuid:2c6b8cb7-8b2a-4051-b291-b1ae6a575ef4 |
| **nodeRepresentation** | Dose form code of the prescribed medicinal product |
| **Code System OID** | 0.4.0.127.0.16.1.1.2.1 |
| **Value** | Textual information describing the Dose form of the prescribed medicinal product |

```
<Classification classificationScheme="urn:uuid:2c6b8cb7-8b2a-4051-b291-
b1ae6a575ef4"
classifiedObject="urn:uuid:343b5a5b-bd3a-49a3-989f-edc1970e059b"
id="urn:uuid:1da2b6bc-aca3-491c-a8a9-526e59f153b0"
nodeRepresentation="50060200">
   <Slot name="codingScheme">
        <ValueList>
            <Value>0.4.0.127.0.16.1.1.2.1</Value>
        </ValueList>
   </Slot>
   <Name>
        <LocalizedString value="Solution for injection in pre-filled pen"/>
   </Name>
</Classification>
```

#### 2.2.2.1.1.6 Prescriber

```
<Classification classificationScheme="urn:uuid:93606bcf-9494-43ec-9b4e-
a7748d1a838d"
classifiedObject="urn:uuid:343b5a5b-bd3a-49a3-989f-edc1970e059b"
id="urn:uuid:c10a9fef-3855-429c-a27a-ada4b966e35d"
nodeRepresentation="">
    <Slot name="authorPerson">
        <ValueList>
            <Value>DIEGO PUENTE GONZALEZ</Value>
        </ValueList>
   </Slot>
</Classification>
```

#### 2.2.2.1.1.1 Substitution

| | |
|---|---|
| **classificationCode** | urn:uuid:2c6b8cb7-8b2a-4051-b291-b1ae6a575ef4 |
| **nodeRepresentation** | eHDSI Substitution code |
| **Code System OID** | 2.16.840.1.113883.5.1070 |
| **Value** | Display name for the substitution code |

```
<Classification classificationScheme="urn:uuid:2c6b8cb7-8b2a-4051-b291-
b1ae6a575ef4"
classifiedObject="urn:uuid:5e4f66e0-9571-43be-a48a-9f786e4a0891"
id="urn:uuid:30a1debb-384d-4077-883a-a3e0c3e9450e"
nodeRepresentation="N">
    <Slot name="codingScheme">
        <ValueList>
```

```
            <Value>2.16.840.1.113883.5.1070</Value>
        </ValueList>
    </Slot>
    <Name>
        <LocalizedString value="none"/>
    </Name>
</Classification>
```

### 2.2.3 Error Messages

If the *eHealth DSI Fetch Data Service* provider is unable to successfully process the query request it MUST respond with a ListResponse message that only contains a <AdhocQueryResponse/RegistryResponse> element.

The following table contains the errors that can be thrown in response to an XCA Query request.

| Type | Code | Description |
|---|---|---|
| Insufficient rights | 4703 | Either the security policy of country A or a privacy policy of the patient (that was given in Country A) does not allow the requested operation to be performed by the Health Provider |
| Generic Security Error | ERROR_SEC_GENERIC | Security validations for the request or the requestor failed |
| Invalid field Error | ERROR_INCORRECT_FORMATTING | The request contains a syntax error |
| Missing field Error | ERROR_MISSING_REQUIRED_FIELDS | The request is missing required information that is necessary to process the request |
| No ePrescription Found Error | ERROR_EP_NOT_FOUND | No ePrescription is registered for the given patient |
| No Patient Summary Found Error | ERROR_PS_NOT_FOUND | No Patient Summary was found for the given patient |
| No ORCD Found Error | ERROR_ORCD_NOT_FOUND | No ORCD document is registered fort he given patient |
| No Patient Consent Error | ERROR_NO_CONSENT | The patient has not given consent to the service |
| Generic Service Signifier Unknown Error | ERROR_GENERIC_SERVICE_SIGNIFIER_UNKNOWN | Class code not supported |
| ePrescription Registry Not Accessible Error | ERROR_REGISTRY_NOT_AVAILABLE | ePrescription registry could not be accessed |
| ORCD Registry Not Accessible Error | ERROR_ORCD_GENERIC | orCD registry could not be accessed |

For more information see [Technical Annexes]

## 2.3 Expected Actions

On a successful FindDocuments query NCP-B MUST take one of the following actions:

- If no matching document was discovered: Terminate the *FetchDocument* service handing back a success indicator to the original requestor

- If matching documents were discovered: Issue a *CrossGatewayRetrieve* request for all documents that were discovered by the *FindDocuments* query

- If matching documents were discovered: Return control to HP-B or any country-B service for selecting among the discovered documents. After a choice on the documents has been done, control MUST be given back to the *FetchDocuments* service which then issues a *CrossGatewayRetrieve* request for all selected documents

In case of an error during document discovery the Fetch Data service gives control back to the issuer of the original request. An error code that indicated the cause of the failure SHOULD be provided to the requestor.

## 2.4 CrossGatewayRetrieve Messages

For retrieving the requested documents eHealth DSI *FetchDocument* service MUST implement the IHE *CrossGatewayRetrieve* message as specified in [IHE ITI TF-ITI-39] XCA.

### 2.4.1 Example Response Messages

See [Technical Annexes]

# 3 eHDSI Compatibility Profile for Overview Documents

## 3.1 Find Documents Request Message

The Find Documents Query request message MUST follow the constraints as defined in section 2.1 of this document. Semantic signifiers on overview documents MUST NOT be used.

## 3.2 Find Documents Response Message

The Find Documents Query response message MUST follow the constraints as defined in section 2.2 of this document. The provided metadata MAY contain additional information about the requested document. This information MUST be encoded as a string and contained within the document metadata comments attribute.

## 3.3 Expected Actions

On a successful FindDocuments query NCP-B MUST take one of the following actions:

- If no matching document was discovered: Terminate the FetchDocument service handing back a success indicator to the original requestor
- If matching documents were discovered: Render the requested overview document from the provided metadata. Return control to HP-B or any country-B service for displaying the overview document to the user.

In case of an error during document discovery the Fetch Data service gives control back to the issuer of the original request. An error code that indicates the cause of the failure SHOULD be provided to the requestor.

# 4 Security Considerations

## 4.1 Application of eHealth DSI Security Services

Implementations of this binding MUST consider the eHealth DSI standard security safeguards in order to preserve basic protection for data confidentiality, data integrity and patient privacy:

- All messages MUST make use of the Web Service Security Standard and the WS Addressing Standard as defined in [Messaging Profile]
- Each request message MUST contain claims on the identity of the requestor and on the care context as defined in [SAML Profile] and [Messaging Profile]

- The service provider MUST verify the integrity and authenticity of these claims for each incoming request
- Before accepting provided data the service provider MUST verify the existence of a valid consent and assess all applicable security policies

## 4.2 XCA Specific Security Measures

### 4.2.1 Patient Identifier

The eHealth DSI functional specification requests that a patient identifier is provided with the Fetch Documents request. While this can be easily mapped onto the IHE XCA Cross-Gateway Query arguments, the IHE XCA Cross-Gateway Retrieve transaction does not provide a slot for including a patient id. Implementations of this binding therefore

- MUST consider the patient identifier as provided within the TRC assertion for the Cross-Gateway Retrieve transaction
- MUST verify that the patient identifier given in the initial request equals the patient identifier given in the TRC assertion. In case of a mismatch, an error MUST be returned to the requestor.

### 4.2.2 Document ID Guessing

An misbehaving user MAY try to access data without authorization by sending a Cross-Gateway Retrieve requests with a guessed or known document identifier of a document that belongs to a patient who has not authorized this HP. In case of a valid Identity Assertion and a valid TRC assertion this misbehaviours is only detected through the assessment of audit trail records.
In order prevent such misbehaviour actively, implementations of this binding:

- MUST verify that all documents provided by NCP-A are assigned to the patient who is indicated in the TRC assertion. In addition to the patient information, extra parameters related to the document identification MIGHT be provided (document ID, PIN code etc.) according the [SAML Profile]. **In case that one or more of the requested documents are assigned to another patient, NCP-A**
    - MUST NOT provide any documents to NCP-B
    - MUST respond with a "No Data" warning in order to not disclose any information on the existence of a certain document to a potential attacker
    - MUST write a respective audit trail
    - MUST issue a security alert to the operator of NCP-A. The operator of NCP-A MUST take responsibility that the alert is as well made known to the operator of NCP-B
    - SHOULD block all further requests that are issued by the suspect HP

## 4.3 Security Audit Considerations

The service consumer MUST write an audit trail entry according to the *HP Assurance Audit Schema* as defined in [Audit Trail Profiles]. The service provider MUST write an audit trail entry according to the *Patient Privacy Audit Schema* as defined in [Audit Trail Profiles].
The following table defines which categories MUST be filled (R), which MAY be filled (O) and which categories MUST NOT be used (X).

| eHealth DSI Instance | Opt. | Description |
|---|---|---|
| Event | R | Audited event |
| Requesting Point of Care | R / X | HCPO that issued the original request. This category MUST be filled by the service consumer. It MUST NOT be provided by the service provider. |
| Human Requestor | R | HP that triggered the request |

| | | |
|---|---|---|
| Source Gateway | R | Service consumer node address at the country of care |
| Target Gateway | R | Service provider node address at the country of the patient's affiliation |
| Audit Source | R | Legal entity that ensures the uniqueness of the identifiers that are used to identify active participants |
| Patient | R | Patient |
| Event Target | R | References to the provided documents (see below) |
| Error Message | O | Only used in case that the request handling was not completed successfully or for anomaly detection |

For the Event Target Category the following fields MUST be provided:

| Field Name | Opt. | Value Constraints |
|---|---|---|
| ParticipantObjectTypeCode | R | MUST be "2" (System Object) |
| ParticipantObjectTypeCodeRole | R | MUST be "4" (Resource) |
| ParticipantObjectIDTypeCode | R | MUST be "12" (URI) |
| ParticipantObjectID | R | MUST be string-encoded UIDs of the provided documents |

# 5 References


[IHE ITI ODD] IHE International: *IT Infrastructure Technical Framework Supplement On-Demand Documents Trial Implementation*, August 2010, accessible under: http://www.ihe.net/Technical_Framework/upload/IHE_ITI_Suppl_On_Demand_Documents_Rev1-1_TI_2010-08-10.pdf

[IHE ITI TF-1] IHE International*: IT Infrastructure Technical Framework Volume 1 – Integration Profiles*. Revision 19, June 2022. https://profiles.ihe.net/ITI/TF/Volume1/index.html

[IHE ITI TF-2] IHE International: IHE IT Infrastructure (ITI) Technical Framework. Volume 2: Transactions. Revision 19, June 2022. https://profiles.ihe.net/ITI/TF/Volume2/index.html

[IHE ITI TF-ITI-38] IHE International*: IT Infrastructure Technical Framework Volume 2 – Transaction ITI-38.* Revision 19, June 2022. https://profiles.ihe.net/ITI/TF/Volume2/ITI-38.html

[IHE ITI TF-ITI-39] IHE International*: IT Infrastructure Technical Framework Volume 2 – Transaction ITI-39.* Revision 19, June 2022. https://profiles.ihe.net/ITI/TF/Volume2/ITI-39.html

[IHE ITI TF-3] IHE International: IHE IT Infrastructure (ITI) Technical Framework. Volume 3 – *Integration Profiles*. Revision 19, June 2022. https://profiles.ihe.net/ITI/TF/Volume3/index.html

[RFC 2119] Bradner, S.: Key words for use in RFCs to Indicate Requirement Levels. Harvard University, Boston, Massachusetts, 1997.